

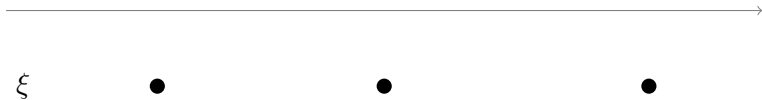
Growth of supercritical Crump-Mode-Jagers Processes without Malthusian parameter

Jakob Stonner

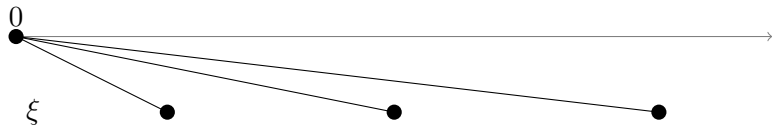
Justus-Liebig-Universität Gießen

Discrete Random Structures II

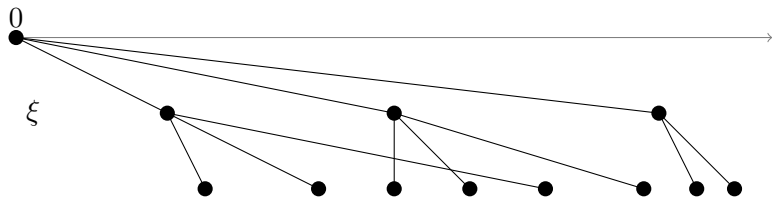
Crump-Mode-Jagers Processes



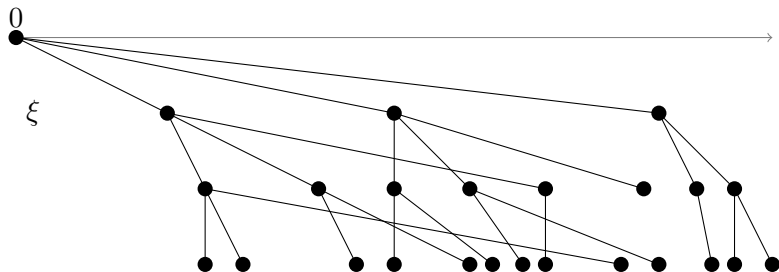
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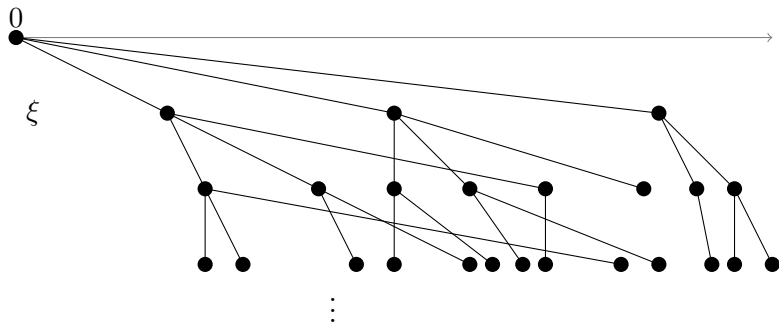
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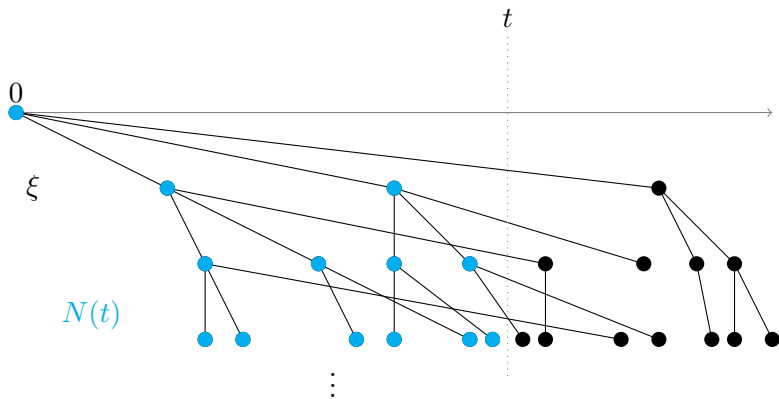
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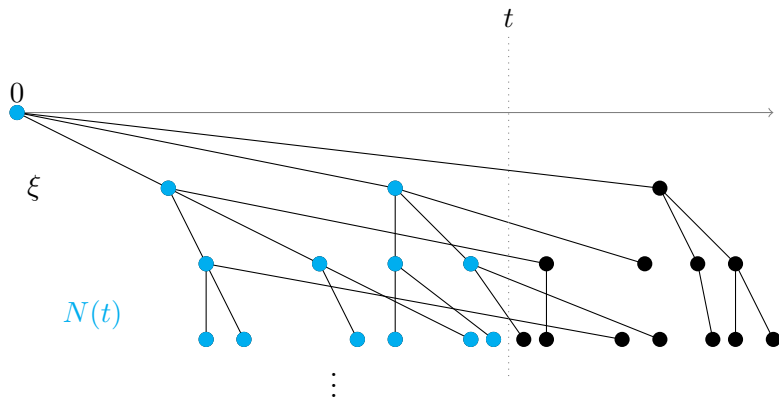
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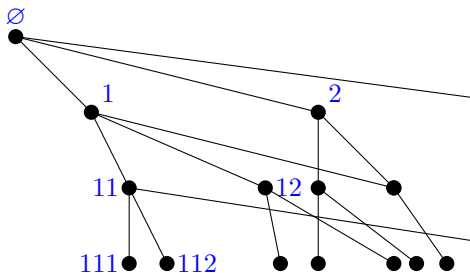
Crump-Mode-Jagers Processes



Question: How fast grows $N(t)$ as $t \rightarrow \infty$?

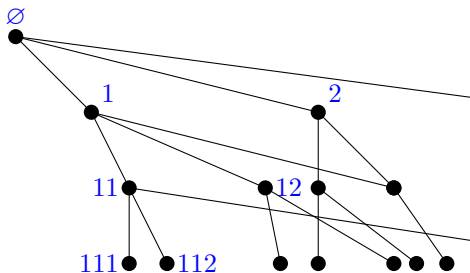
CMJ process definition

Let $\mathcal{I} := \bigcup_{n \in \mathbb{N}_0} \mathbb{N}^n$



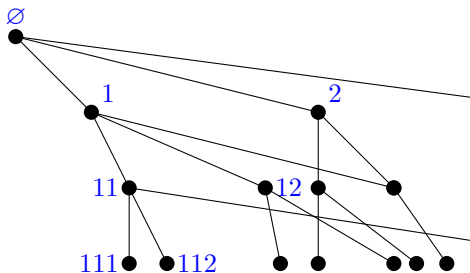
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 $(\sum_j \delta_{X_j(u)})_{u \in \mathcal{I}}$ i.i.d. $\sim \xi$.



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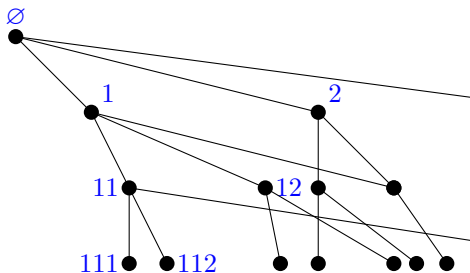
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$$S(\emptyset) := 0, \quad S(uj) := S(u) + X_j(u), \quad u \in \mathcal{I}, j \in \mathbb{N}$$

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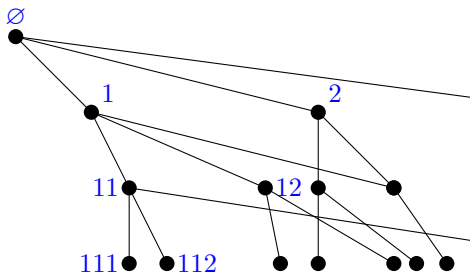


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Example: $\phi = \mathbb{1}_{[0, \infty)} \Rightarrow \mathcal{Z}^\phi(t) = N(t)$

How fast does $\mathcal{Z}^\phi(t)$ grow?

Let

$$m(\theta) := \mathbb{E} \left[\int e^{-\theta x} \xi(\mathrm{d}x) \right], \quad \theta \geq 0.$$

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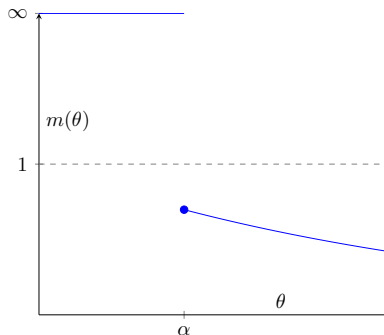
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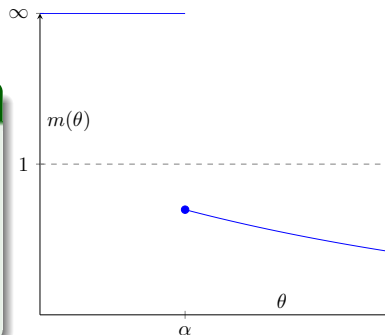
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Example

Poisson point process with intensity $e^x f(x) \mathrm{d}x$, f heavy-tailed, $\int f(x) \mathrm{d}x < 1$ (e.g. $f(x) = x^{-p} \ell(x)$)

$$m(\theta) = \int e^{(1-\theta)x} f(x) \mathrm{d}x$$



Mean asymptotics

Assume

$$m(1) < 1, \quad \forall \theta < 1 : m(\theta) = \infty.$$

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Let $F(dx) := e^{-x} \mathbb{E}[\xi(dx)] \Rightarrow$ renewal equation

$$e^{-t} \mathbb{E}[\mathcal{Z}^\phi(t)] = e^{-t} \mathbb{E}[\phi(t)] + \int e^{-(t-x)} \mathbb{E}[\mathcal{Z}^\phi(t-x)] F(dx)$$

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Proof: apply [Asmussen, Foss, Korshunov 2003]

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Then we have

$$\frac{e^{-t}}{g(t)} \mathcal{Z}^\phi(t) \xrightarrow[t \rightarrow \infty]{\mathbb{P}} \sum_{u \in \mathcal{I}} e^{-S(u)} \Phi_u + \frac{\int_0^\infty e^{-x} \mathbb{E}[\phi(x)] dx}{1 - m(1)} \sum_{u \in \mathcal{I}} e^{-S(u)} L_u.$$

Implications and Examples

Corollary

Assume F has a subexponential asymptotic density g and $\frac{1}{g(t)} \int_{(t,t+T]} e^{-x} \xi(dx) \xrightarrow[t \rightarrow \infty]{\mathbb{P}} TL$ for all $T > 0$. Then

$$\frac{e^{-t}}{g(t)} N(t) \xrightarrow[t \rightarrow \infty]{\mathbb{P}} \frac{1}{1 - m(1)} \sum_{u \in \mathcal{I}} e^{-S(u)} L_u.$$

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 $\Rightarrow L = 1$
- $\xi' = \theta_X \xi$, $X \geq 0 \Rightarrow m'(\theta) = \mathbb{E}[e^{-\theta X}] m(\theta)$, $L' = e^{-X} L$.

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Seneta-Heyde scaling for the intrinsic martingale

For $\mathcal{C}_t := \{uj \in \mathcal{I} : S(u) \leq t < S(uj)\}$,

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Outlook

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Thanks for listening!